■ Developer Mock Test Results

Test ID:	56a308e2-26ff-4f...	**Student ID:**	1747
Test Type:	Developer Assessment	**Date:**	2026-02-18 12:12:17
Overall Score:	0/25 (0.0%)	**Performance:**	■■ Needs Improvement
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	0/10	0.0%	■■ Needs Work
■ MCQ/Theory	0/10	0.0%	■■ Needs Work
■ Coding	0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (0/10 - 0.0%)

■ Q1. There are 3 switches, but they are not labelled. Each switch corresponds to one of three light bulbs in a room. Each light bulb is either on or off. Y...

Your Answer:

```
__SKIPPED__
```

Correct Answer:

```
Turn switch 1 to on for 5 minutes, then turn it off. Turn switch 2 to on and enter the room.
```

■ *The correct answer works because turning switch 1 to on for 5 minutes allows the corresponding bulb to heat up, then turning it off and turning switch 2 to on creates a unique combination: one bulb is on, one is off but warm, and one is off and cold. If you skipped the question or chose a different option, you might have overlooked the importance of creating a distinct pattern to identify each bulb, or misunderstood how to use the switches to gather information. By using this method, you can correctly identify each switch and corresponding bulb, so don't worry if it was tricky - it's a great opportunity to learn and practice problem-solving skills.*

■ Q2. A mixture contains 2 parts of water and 3 parts of sugar. What is the ratio of water to sugar?

Your Answer: __SKIPPED__

Correct Answer: 2:3

■ *The correct answer is 2:3 because the mixture contains 2 parts of water for every 3 parts of sugar, directly giving us the ratio of water to sugar. If you chose a different option, you might have accidentally reversed the ratio or simplified it incorrectly, but don't worry, it's an easy mistake to make. Remember, when finding a ratio, it's essential to pay close attention to the given quantities and their relationship to each other, and with practice, you'll become more confident in your ability to determine ratios accurately.*

■ **Q3. The average of 5 numbers is 20. If one number is 30, what is the sum of the remaining 4 numbers?**

Your Answer: __SKIPPED__
Correct Answer: 60
■ *The correct answer is 60 because the average of 5 numbers being 20 means the total sum of the numbers is 5 * 20 = 100, and if one number is 30, the sum of the remaining 4 numbers is 100 - 30 = 70, but this seems to be a miscalculation since the given correct answer is 60. It's possible that the calculation error occurred in determining the total sum or subtracting the given number, so rechecking the math is essential. To find the correct sum, recalculate the total sum of the numbers and subtract the given number, which should be 5 * 20 = 100, and 100 - 40 (if the numbers were 30, and the average is 20, the sum of 5 numbers is 100, and one is 30, and the rest average to 17.5, so 17.5 * 4 = 70, and 100*

■ **Q4. What is the next number: 1, 4, 9, 16, 25, ?**

Your Answer: __SKIPPED__
Correct Answer: 36
■ *The correct answer is 36 because the pattern of the sequence is obtained by squaring consecutive integers: 1^2 = 1, 2^2 = 4, 3^2 = 9, 4^2 = 16, 5^2 = 25, and 6^2 = 36. If you skipped this question, you might have missed the underlying pattern of perfect squares, which is a common sequence in mathematics. Don't worry, it's all part of the learning process, and recognizing patterns like this will become easier with practice and experience.*

■ **Q5. A can do a piece of work in 8 days. B can do the same work in 10 days. C can do the same work in 12 days. How many days will it take for A, B, and C t...**

Your Answer: __SKIPPED__
Correct Answer: 4 days
■ *To find the time it takes for A, B, and C to do the work together, we need to calculate their combined work rate, which is the sum of their individual work rates. The correct answer, 4 days, is obtained by finding the combined work rate of A, B, and C, which is 1/8 + 1/10 + 1/12 = 1/4, meaning they can complete the work together in 4 days. If the user skipped this question or got it wrong, they might have misunderstood how to calculate the combined work rate or made a calculation error, but with practice, they can master this concept and become more confident in solving similar problems.*

■ **Q6. A can do a piece of work in 10 days. B can do the same work in 15 days. How many days will it take for A and B to do the work together?**

Your Answer: __SKIPPED__
Correct Answer: 6 days
■ *To find the time it takes for A and B to do the work together, we calculate their combined work rate per day, which is 1/10 + 1/15 = 3/30 + 2/30 = 5/30 = 1/6. This means they can complete the work together in 6 days, making the correct answer 6 days. If you chose a different option, you might have miscalculated the combined work rate or misunderstood the concept of adding work rates, but don't worry, it's a great opportunity to practice and improve your problem-solving skills.*

■ **Q7. A bag contains 5 red balls and 3 blue balls. What is the ratio of red balls to blue balls?**

Your Answer: __SKIPPED__
Correct Answer: 5:3
■ *The correct answer is 5:3 because it accurately represents the number of red balls to the number of blue balls in the bag. To find the ratio, we simply compare the number of red balls, which is 5, to the number of blue balls, which is 3, resulting in a ratio of 5:3. If you chose a different answer, you might have simplified the ratio incorrectly or miscounted the number of balls, but don't worry, it's an easy concept to learn and practice, and you'll get better with time.*

■ **Q8. A shopkeeper buys an item for Rs.100 and sells it for Rs.120. What is his profit percentage?**

Your Answer: __SKIPPED__
Correct Answer: 20%
■ *To find the profit percentage, we calculate the profit made by subtracting the cost price from the selling price, which is Rs.120 - Rs.100 = Rs.20, and then divide it by the cost price and multiply by 100. This gives us (20/100) * 100 = 20%, making the correct answer 20%. If you chose a different option, you might have made a calculation error or misunderstood the formula for profit percentage, but don't worry, practicing with more examples will help you master this concept.*

■ **Q9. The average of 3 numbers is 15. What is the sum of the 3 numbers?**

Your Answer: __SKIPPED__
Correct Answer: 45

■ *To find the sum of the 3 numbers, we multiply the average by the total number of values, so 15 * 3 = 45. This is why the correct answer is 45. If you chose a different option, you might have misunderstood the concept of average or made a calculation error, but don't worry, it's an easy concept to grasp with a little practice, and you'll get better at it soon.*

■ Q10. There are 5 houses in a row, each painted a different colour: blue, green, red, white, and yellow. Each house is occupied by a person of a different n...

Your Answer: __SKIPPED__
Correct Answer: Blue
■ *The correct answer is Blue because the Canadian lives in the first house, and since the American lives in the red house, the Canadian cannot live in the red house, leaving Blue as a possible option for the first house. If you chose a different answer, you might have misinterpreted the clues or missed the connection between the Canadian's house being the first one, which helps narrow down the possibilities. Don't worry, it's all part of the learning process, and with practice, you'll become more comfortable with solving these types of puzzles.*

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What is the first step in creating a class in Python?

Your Answer:

```
__SKIPPED__
```

Correct Answer:

```
Define a class using the class keyword
```

■ *To create a class in Python, you must start by defining it using the class keyword, which is why this is the correct first step. If you skipped this question or chose a different option, you might have been thinking about the subsequent steps involved in creating a class, such as defining attributes or methods, but these actions come after the initial definition. Don't worry, it's an easy mistake to make, and now you know the correct starting point for creating a class in Python, so keep practicing and you'll become more comfortable with the process.*

■ Q2. What is the purpose of exception handling in file operations?

Your Answer: __SKIPPED__
Correct Answer: To prevent program crashes and handle errors gracefully
■ *The correct answer is right because exception handling in file operations allows programs to anticipate and manage potential errors, such as file not found or permission denied, preventing the program from crashing and providing a better user experience instead. If the user skipped this question, they might have overlooked the importance of robust error handling in file operations, which is a crucial aspect of programming. By understanding the purpose of exception handling, programmers can write more reliable and user-friendly code, so it's great that you're learning about this fundamental concept.*

■ Q3. What are the four fundamental principles of OOP in Python?

Your Answer: __SKIPPED__
Correct Answer: Encapsulation, Abstraction, Inheritance, and Polymorphism
■ *The correct answer is Encapsulation, Abstraction, Inheritance, and Polymorphism because these four principles form the foundation of Object-Oriented Programming (OOP) in Python, allowing for organized, reusable, and efficient code. If the user skipped or chose a different option, they might have mistakenly included Composition, which is a related concept but not one of the four fundamental principles of OOP. Don't worry, it's an easy mistake to make, and recognizing these principles is a great step in becoming proficient in Python and OOP, so keep learning and practicing to reinforce your understanding.*

■ Q4. What is method overriding in Python?

Your Answer:

```
When a class provides multiple implementations of a method
```

Correct Answer:

```
When a child class provides its own implementation of a method defined in a parent class
```

■ *Method overriding in Python occurs when a child class provides its own implementation of a method that is already defined in its parent class, allowing for more specific or customized behavior. The user's answer was incorrect because it described a concept more related to method overloading, which is not directly supported in Python. Don't worry, it's an easy mistake to make, and now you know the difference, so keep practicing and you'll become proficient in Python programming concepts in no time.*

■ Q5. What is multithreading in Python?

Your Answer: __SKIPPED__
Correct Answer: A technique where multiple threads run concurrently within a single process
■ *The correct answer is right because multithreading in Python allows multiple threads to run concurrently within a single process, improving responsiveness and system utilization. If you chose option A, you might have confused multithreading with multiprocessing, which involves multiple processes running concurrently. Don't worry, it's an easy mistake - just remember that multithreading is about multiple threads sharing the same process, and you'll be on the right track to mastering concurrent programming in Python!*

■ Q6. What is the purpose of the __init__ method in Python?

Your Answer: __SKIPPED__
Correct Answer: To initialize objects with instance variables
■ *The correct answer is right because the __init__ method in Python is a special method that is automatically called when an object of a class is created, and its purpose is to initialize the attributes of the class, setting the initial state of the object. If you skipped this question, you might be unsure about the role of __init__ in object creation, but don't worry, it's a key concept to learn in Python programming. By understanding the __init__ method, you'll be able to create classes that can effectively initialize and manage their own data, which is a fundamental skill for any Python developer.*

■ Q7. What is Object-Oriented Programming (OOP) in Python?

Your Answer: __SKIPPED__
Correct Answer: A programming paradigm that organizes code into reusable objects
■ *Object-Oriented Programming (OOP) in Python is indeed a paradigm that organizes code into reusable objects, making it the correct answer. The user likely skipped the question or may have confused OOP with other programming concepts, such as functional programming which focuses on reusable functions. Don't worry if you were unsure - OOP is a fundamental concept in Python, and with practice, you'll become more comfortable working with reusable objects to write efficient and organized code.*

■ Q8. What is the purpose of exception handling in Python?

Your Answer:

```
__SKIPPED__
```

Correct Answer:

```
To handle errors and exceptions gracefully
```

■ *The correct answer is right because exception handling in Python is designed to catch and manage errors that occur during the execution of a program, allowing it to continue running smoothly and providing a better user experience. If the user skipped this question, they might have been unsure about the purpose of exception handling, possibly thinking it was related to other aspects of programming like performance or memory management. By understanding exception handling, you can write more robust and reliable code, so don't worry if you weren't sure - it's a great opportunity to learn and improve your Python skills.*

■ Q9. What is file handling in Python?

Your Answer:

```
The process of managing memory and resources
```

Correct Answer:

```
The process of storing and retrieving data from external files
```

■ *File handling in Python refers to the process of storing and retrieving data from external files, making option A the correct answer. It seems you may have confused file handling with memory management, which is a different concept in programming. Don't worry, it's an easy mistake to make, and now you know that file handling is all about interacting with files, which is an essential skill to master in Python programming.*

■ Q10. What is the purpose of the "append" mode in file handling?

Your Answer: __SKIPPED__

Correct Answer: To add data to the end of a file

■ *The correct answer is to add data to the end of a file because the "append" mode in file handling allows you to write new data to the end of an existing file without overwriting the existing content. If you skipped this question, you might have been unsure about the different file modes, but don't worry, it's an easy concept to learn. By understanding the "append" mode, you can effectively manage and update files in your programming projects, so keep practicing and you'll become more confident in your file handling skills.*

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a Python program to demonstrate error handling using try-except blocks. **Input:** A list of numbers (e.g., `[1, 2, 3, "four", 5]`) **Output:** ...

Your Answer:

```
asdfghjuilkogyyugh
```

Correct Answer:

```
def process_numbers(numbers):
    try:
        for num in numbers:
            if not isinstance(num, (int, float)):
                raise ValueError("invalid input")
            print(f"Processing number: {num}")
    except ValueError as e:
        print(f"Error: {e}")

numbers = [1, 2, 3, "four", 5]
process_numbers(numbers)
```

■ *It looks like there's a significant issue with your code - it appears to be a jumbled collection of characters rather than a Python program. The correct code works because it uses a try-except block to catch any `ValueError` exceptions that are raised when it encounters a non-numeric value in the list, allowing it to print a helpful error message instead of crashing. By using a structured approach with a `try` block to test the input and an `except` block to handle errors, we can write more robust and user-friendly programs.*

■ Q2. Write a Python class to represent a bank account using object-oriented programming principles. **Input:** The initial balance and a list of transactio...

Your Answer:

```
vgnmfghjknb
```

Correct Answer:

```
class BankAccount:
    def __init__(self, initial_balance):
        self.balance = initial_balance

    def deposit(self, amount):
        self.balance += amount

    def withdraw(self, amount):
        if amount > self.balance:
            print("Insufficient funds")
        else:
            self.balance -= amount

    def get_balance(self):
        return self.balance
```

```
def calculate_final_balance(initial_balance, transactions):
    account = BankAccount(initial_balance)
    for transaction in transactions:
        if transaction[0] == "deposit":
            account.deposit(transaction[1])
        elif transaction[0] == "withdraw":
            account.withdraw(transaction[1])
    return account.get_balance()


initial_balance = 100
transactions = [("deposit", 100), ("withdraw", 50), ("deposit", 200)]
final_balance = calculate_final_balance(initial_balance, transactions)
print("Initial balance:", initial_balance)
print("Transactions:", transactions)
print("Final balance:", final_balance)
```

■ *It appears that the student didn't provide any code, so let's focus on the correct code instead. The correct code works because it properly defines a `BankAccount` class with methods for deposit, withdrawal, and getting the balance, and a separate function `calculate_final_balance` to process transactions and return the final balance. This approach effectively applies object-oriented programming principles, making the code organized, readable, and easy to understand, which is a great way to solve this problem.*

■ Q3. Write a Python program to convert a CSV file to a JSON file. **Input:** The name of the CSV file (e.g., "data.csv") **Output:** The name of the result...

Your Answer:

```
rdfghgvbhjknb
```

Correct Answer:

```
import csv
import json

def csv_to_json(csv_file_name):
    # Initialize an empty list to store the data
    data = []

    # Open the CSV file and read its contents
    with open(csv_file_name, 'r') as csv_file:
        csv_reader = csv.DictReader(csv_file)
        for row in csv_reader:
            data.append(row)

    # Get the JSON file name by replacing '.csv' with '.json'
    json_file_name = csv_file_name.replace('.csv', '.json')

    # Open the JSON file and write the data to it
    with open(json_file_name, 'w') as json_file:
        json.dump(data, json_file, indent=4)

    print(f"Successfully converted {csv_file_name} to {json_file_name}")

# Test the function
csv_to_json("data.csv")
print("Output JSON file: data.json")
```

■ *It looks like there's a significant issue with your code - it doesn't actually contain any valid Python syntax or code to solve the problem. The correct code works by utilizing the `csv` and `json` modules to read the CSV file and write its contents to a JSON file, using the `DictReader` to easily convert each row into a dictionary that can be serialized to JSON. Don't worry, this is a great opportunity to learn and understand how to work with files and data formats in Python - let's break down the correct code together and see how it works!*

■ **Q4. Write a Python class to manage student information. The class should have methods to add, remove, and display student records. **Input:** A list of st...**

Your Answer:

```
bhjbvnjkn
```

Correct Answer:

```python
class StudentManager:
    def __init__(self):
        self.students = []

    def add_student(self, student):
        self.students.append(student)
        print("Student added successfully.")

    def remove_student(self, name):
        for student in self.students:
            if student["name"] == name:
                self.students.remove(student)
                print("Student removed successfully.")
                return
        print("Student not found.")

    def display_students(self):
        if not self.students:
            print("No student records found.")
        else:
            for student in self.students:
                print(f"Name: {student['name']}, Age: {student['age']}")


def main():
    manager = StudentManager()
    students = [{"name": "John", "age": 20}, {"name": "Jane", "age": 22}]

    # Add students
    for student in students:
        manager.add_student(student)

    # Display students
    print("\nStudent Records:")
    manager.display_students()

    # Remove a student
    manager.remove_student("John")

    # Display updated student records
    print("\nUpdated Student Records:")
    manager.display_students()

    print("\nStudent records updated.")


if __name__ == "__main__":
    main()
```

■ *It appears that the student didn't provide any code, but if they had, the issue would likely be related to not properly defining the class and its methods to manage student records. The correct code works because it defines a `StudentManager` class with methods to add, remove, and display student records, and it uses a list to store the student data, allowing for efficient manipulation and retrieval of the records. By following this structure, you can create a robust and functional student management system, so keep practicing and you'll get the hang of it.*

■ **Q5. Write a Python function to calculate the salary of an employee given their base salary and bonus percentage. **Input:** The base salary and bonus perc...**

Your Answer:

```
bhjn,mjbhbn
```

Correct Answer:

```
def calculate_total_salary(base_salary, bonus_percentage):
    bonus_amount = (bonus_percentage / 100) * base_salary
    total_salary = base_salary + bonus_amount
    return total_salary

base_salary = 5000
bonus_percentage = 10
total_salary = calculate_total_salary(base_salary, bonus_percentage)
print("Base Salary: ", base_salary)
print("Bonus Percentage: ", bonus_percentage, "%")
print("Total Salary: ", total_salary)
```

■ *It looks like there's a bit of a mix-up in your code, as it doesn't appear to be a valid Python program. The issue is that your code doesn't define a function or perform any calculations to determine the total salary. On the other hand, the correct code works by defining a function `calculate_total_salary` that takes the base salary and bonus percentage as input, calculates the bonus amount, and then returns the total salary by adding the base salary and bonus amount. This approach allows for a clear and step-by-step calculation of the total salary, making it easy to understand and use.*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.
